Flutter Take-Home Assignment

Estimated Time: 4 hours (you can take 1-2 days to come back)

Objective

Build a small shopping application using Flutter. The goal of this assignment is to demonstrate your ability to:

- Consume REST APIs
- Manage application state
- Structure and organize code
- Handle edge cases and failures
- Build a polished user experience

API

Use the following API: https://fakestoreapi.com/products

Endpoints

- **Get all products** GET /products
- **Get product details** GET /products/{id}

Required Features

1. Product Listing Screen

Display a list of products fetched from the API.

Each product item should display:

- Product image
- Product title
- Product price
- Product rating

Functionality

- Fetch products from the API when the app launches

- Pull-to-refresh support
- Search products by title
- Loading state while data is being fetched
- Empty state when no products match the search criteria
- Error state when API calls fail
- Retry functionality from the error state

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2. Product Details Screen

Selecting a product should navigate to a dedicated details screen.

Display:

- Product image
- Product title
- Product description
- Product category
- Product price
- Product rating

Functionality

- Add product to favorites
- Remove product from favorites
- Favorite status should be reflected consistently throughout the application

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3. Favorites Screen

Create a separate screen that displays all favorited products.

Functionality

- View all favorite products
- Remove products from favorites
- Favorite products should persist after restarting the application

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Technical Requirements

- Use Flutter and Dart
- Consume data from the provided API

- Manage application state appropriately
- Handle loading, success, empty, and error states
- Persist favorite products locally
- Ensure the application runs on both Android and iOS
- Write clean, readable, and maintainable code
- Use packages only when they provide clear value

Expected User Experience

The application should feel complete and polished.

Users should be able to:

- Browse products immediately after launch
- Search products easily
- View detailed product information
- Add and remove favorites seamlessly
- Reopen the application and retain favorite selections
- Recover gracefully from network failures

Bonus (Optional)

Any of the following:

- Product sorting (price, rating, etc.)
- Additional UX improvements

Submission

Please share:

- Source code repository
- Instructions to run the application
- Any assumptions or trade-offs made during development